



Darwin's Four Postulates in Light of "Don't Die"

By **Neel Somani**

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For most of history, more people meant more power. I came across this X post that I initially found alarming for Western civilization:



 **Sergey Radchenko**  @DrRadchenko · Sep 5

Merkel recounts her meetings with Xi.

his predecessor, Hu, as a matter of courtesy, discussing all the important bilateral matters with Premier Wen Jiabao, once Xi came into power I discussed almost all issues with him. He repositioned China. At our meetings, he spoke frequently about the history of humanity over the past two thousand years and emphasized that, in eighteen of those twenty centuries, China had been the economic and cultural center of the world. China had only fallen behind in the early nineteenth century, he told me. When I heard that, I asked my economic and fiscal advisor Lars-Hendrik Röller to check these statements against the available economic data. He confirmed Xi's portrayal. In Xi's view, China has returned to its historic normality; he called this

Xi -
Merkel

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Source: [Arnaud Bertrand on X](#)

But what I realized is that the argument is subtly incorrect, because it fails to recognize the important shifts happening.

It's easy to see why a negative birth rate society is concerning to most. In "On the Origin of Species," Darwin presents his famous four postulates argument. He frames these as sufficient conditions for natural selection to occur:

1. Variation among individuals
2. Heritability of that variation
3. Overproduction leading to competition
4. Differential reproductive success

Biologists today often treat these not only as sufficient, but also as necessary. So to a biologist, these conditions make clear why a society with a persistently negative birth rate is unsustainable. Without reproduction, traits cannot be propagated, and the lineage eventually disappears.

But the picture is shifting. Fertility rates are falling globally, yet lifespans are lengthening. Ray Kurzweil makes the popular argument that advances in medicine and technology suggest a possible future in which human life expectancy could increase faster than we age. Bryan Johnson speaks of a future where our top priority is simply “don’t die.”

This raises a question: does Darwin’s argument no longer hold up in a Kurzweilian world where we might not die of old age at all?

From Generations to Timesteps

Darwin’s framing relied on “generations” as the natural unit of evolutionary change. Evolution, he argues, is the process of the gene pool shifting from generation to generation. This was a reflection of his early understanding of biology, but our modern view on evolution has adapted over the last 100 years.

This raises a question: does Darwin’s argument no longer hold up in a Kurzweilian world where we might not die of old age at all? 

Generations are noisy to measure (organisms reproduce at different ages) and the distinction between parents and children is less fundamental than it appears. A “child” is simply a continuation of the biochemical process that ran in its parent, part of a continuous stream of life extending back to the origin of life itself.

The more precise discretization of evolution is therefore “timesteps” (t , $t+1$, $t+2$, ...). Recast in these terms, Darwin’s “four” postulates become:

1. Variation exists at a given timestep
2. Traits or states persist across timesteps (retention)
3. [redacted]

4. Certain traits are differentially favored in persistence or propagation

This version no longer requires reproduction per se, only continuity of adaptive traits across timesteps. And it no longer requires Darwin's third postulate on competition. Instead, the fourth postulate implies some filtering or differential persistence mechanism exists.

How the Negative Birth Rate Society Wins

Reframed in this way, whether a negative birth rate society can be “last standing” depends on several factors:

- How long individuals live
- Whether reproduction is the only vehicle of continuity
- Whether institutions, technologies, or cultural systems can preserve and propagate traits
- Whether gene flow from immigration supplements declining fertility
- What selection mechanisms govern persistence from timestep t to timestep $t+1$

Three scenarios illustrate how this could work:

1. Longevity and Retention

If individuals are extremely long-lived and capable of adapting their traits, reproduction becomes less central. Selection acts through endurance: societies with long-lived, adaptable members could outlast more fertile rivals that collapse under resource strain, internal instability, or external shocks. Here, persistence replaces reproduction as the critical unit of selection.

Selection acts through endurance: societies with long-lived, adaptable members could outlast more fertile rivals... 

The Venetian Republic illustrates this dynamic. It endured for centuries not because of demographic strength but because of institutional resilience.

2. Informational Continuity

What traits are we measuring? If the true “offspring” of a society are not its biological children but its cultural, institutional, and technological systems, then fertility is not the bottleneck. AI, automation, cloning, or cultural transmission can carry traits forward.

The Jewish diaspora is a powerful example. For millennia, Jewish communities preserved identity and continuity without a sovereign state or demographic dominance. In today’s world, Japan, despite its low fertility, remains globally significant due to its cultural, technological, and institutional robustness.

3. Immigration and Emigration

Another way to shape which traits persist is through migration policy. A society can import individuals with desirable traits (immigration) while exporting or excluding those with less adaptive traits (emigration). In the case of a negative birth rate society, forced emigration isn’t required. You can just wait. Declining fertility naturally reduces the prevalence of traits over time.

Singapore demonstrates a modern application. Its fertility is among the lowest in the world, but it deliberately uses selective immigration and stringent residency policies to curate its demographic and cultural profile.

Control the Information Stream

A society with low or negative fertility could still be last standing if it sustains the continuity of its traits better than others, whether through longevity, institutional robustness, or technological surrogates for reproduction. And in some cases, fewer people may even be a strength, as wealth and resources per capita increase. That means more resources per person, and more extra resources for defense.

Survival is no longer about replacing people, but about maintaining the unbroken stream of information and function that defines society itself. 🐦

In this sense, survival is no longer about replacing people, but about maintaining the unbroken stream of information and function that defines society itself. We don't necessarily need solutions to the "reproductive crisis." The challenge for us is to decide what mechanisms we should design to govern the information stream over time, and what kind of continuity we want to preserve.



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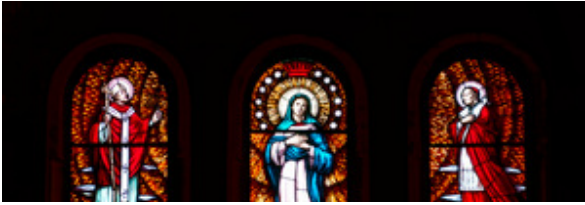
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